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Mobilization of Oil through In Situ Emulsification in a Porous Medium: A Comparative Analysis of Oil Solubilization and Interfacial Tension Reduction

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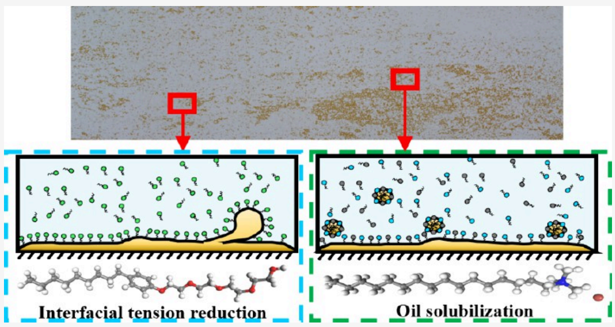

ABSTRACT: The addition of surfactants into the aqueous phase is acknowledged as a primary mechanism for the reduction of the interfacial tension (IFT) between oil and water, providing oil mobilization within confined conditions. Certain surfactants exhibiting ultralow interfacial tension can solubilize oil, thereby generating microemulsions, whereas other surfactants are limited to the formation of thermodynamically unstable emulsions. There is currently no evidence to determine which factor holds greater significance for oil mobilization within porous media where oil and water are present simultaneously. Two surfactants, each exhibiting ultralow interfacial tension, were systematically screened and assessed through a series of experiments, resulting in the formation of middle phase microemulsions and emulsions. The investigation focused on their performance in oil mobilization prior to and following water flooding, utilizing micromodel and core displacement experiments. It was found that cetyltrimethylammonium bromide (CTAB) with *n*-butanol featured good oil solubilization capability and could form microemulsions with a wide carbon number distribution of alkanes. Nonylphenol polyoxyethylene ether carboxylate (SS-231) featured the ultralow IFT with simulated oil and could form dense emulsions. The pore-scale and core-scale oil displacement experiments both showed that the CTAB with *n*-butanol could get a higher ultimate oil recovery than that of SS-231 without prewater flooding, while the results became contrary if the surfactants were injected after water flooding. Our work found that the mechanism of oil solubilization worked well when there was only an oil phase in the porous medium, and the emulsion formation became more important for oil mobilization at the high-water cut stage. The results indicated that the surfactant featured with good emulsification capability was preferred at the high-water cut stage while surfactants featured with high oil−water solubilization capacity could be good candidates if the remaining oil saturation was high in the reservoir.

1. INTRODUCTION

The displacement between two-phase fluids occurs in many natural conditions and engineering processes. Its efficiency depends on the competition of various forces, such as capillary force, viscous force, and gravity.[1−3] To accelerate or restrain fluid peel-off from the solid surface, surfactant is usually added for the alteration of surface and interfacial energies.[4,5] Among its many practical applications, oil recovery from subsurface porous media has garnered significant research interest due to its theoretical value and broad application prospects.[6−8]

Water injection to displace oil is the most commonly used technique for oil production from the reservoir. The capillary force dynamic variation is due to the interfacial curvature alterations of two-phase fluids flowing through the pore throat, which dominate the displacement efficiency of immiscible fluids.[9] The capillary force positively correlated with the interfacial tension of the two-phase fluids and negatively correlated with the pore radius. It acted as the resistance when the nonwet phase displaced the wet phase and vice versa.[10,11]

In the real displacement process, drainage and imbibition always successively occur when fluids enter and exit the pore throat. It indicates that the capillary force would be both a driving and a resistance force, whether on wet phase or nonwet phase fluids flow through the pore throat. Threshold pressure of fluid entry into pores poses a challenge for oil mobilization. Therefore, the surfactant is suggested to be added to reduce the effect of capillary force in the process of oil production.[12,13] However, there are two different viewpoints on the oil mobilization mechanisms by surfactants, which result in a different selection of surfactants.

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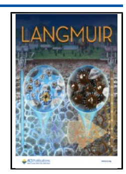

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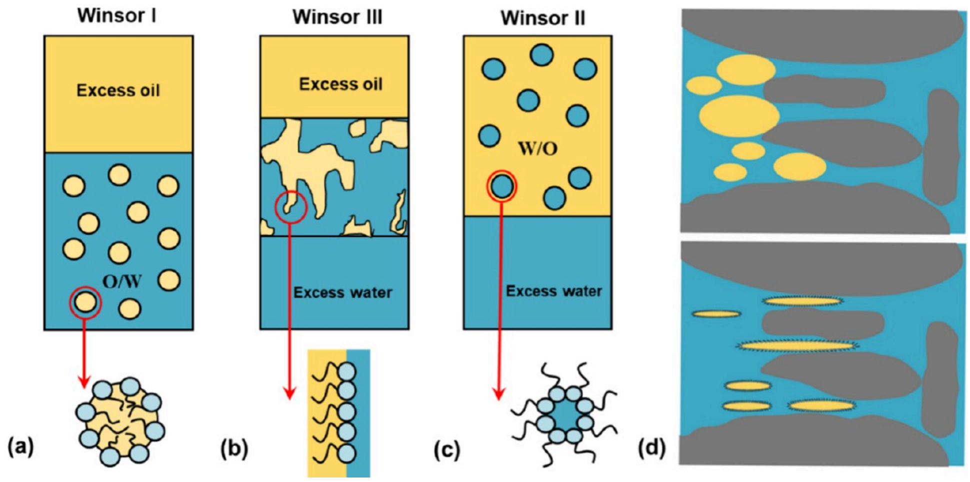

Figure 1. Schematic diagram of two EOR mechanisms of surfactant: (a) Winsor I (lower phase) microemulsion formation by the oil phase solubilized into the aqueous phase, (b) Winsor III (middle phase) microemulsion formation by the surfactant simultaneously solubilizing oil and water, (c) Winsor II (upper phase) microemulsion formation by the aqueous phase solubilized into the oil phase, and (d) trapped oil mobilized by fluid due to the low IFT effect.

The reduction of interfacial tension between oil and water, or the formation of microemulsions through surfactant solubilization, can effectively reduce the effect of capillary forces.[14,15] These two characteristics can occasionally manifest within a single surfactant system. The interfacial tension between oil and water can be determined from the solubilization ratios of both phases, utilizing Huh correlations in the presence of microemulsion formation.[16] The value would be significantly diminished if the solubilization ratio exceeds 10. Microemulsions can be categorized into three distinct types, depending on the surfactant's ability to solubilize oil and water, as shown in Figure 1. The aqueous phase solubilized into the oil phase formed the upper phase microemulsion, and the oil phase solubilized into the aqueous phase formed the lower phase microemulsion with the help of surfactant.[17] The middle phase microemulsion will be formed when the surfactant simultaneously solubilizes oil and water. Compared to the other two types of microemulsions, the middle phase microemulsion presented the lower IFT with both the aqueous phase and the oil phase, which is favorable for oil recovery. Moreover, the equal water and oil solubilization ratio in the middle phase microemulsion featured the lowest IFT between oil and water. Its value can be handled by the salinity changing. Therefore, the optimal salinity, which corresponds to the equal water and oil solubilization ratio, is pursued in the surfactant design or screening toward enhanced oil recovery (EOR).[18,19] However, many surfactants featured ultralow IFT but no microemulsion formation and were successfully applied in surfactant−polymer (SP) flooding and alkali-surfactant−polymer (ASP) flooding pilots.[20−22] The ultra-IFT between surfactant and crude oil measured by the spinning drop interface tensiometer was regarded as a standard for the surfactant screening.[23,24] Oil solubilization ability is the main difference between the phase behavior tests and IFT direct measurements for surfactant screening.

Laboratorial experiments and pilot tests both show that surfactant with the ultralow IFT or microemulsion formation could achieve a significantly higher oil recovery factor, but their differences on the oil mobilization at various oil−water contact conditions are still unclear. Previous successful SP and ASP pilot tests in China both used the diluted petroleum sulfonate surfactants with a concentration of less than 0.3 wt %. Since the phase behavior tests were not performed in the surfactant screening experiments, we only know that the surfactants featured the ultralow IFT with crude oil but are not sure whether they can form microemulsions. The incremental oil recovery factors from 15 to 25% had been achieved in these pilots even though the chemical flooding was initiated at the high-water cut stage.[25−27] Some emulsions, rather than microemulsions, were observed in parts of the pilots' effluent.[28,29] It is hypothesized that the absence of microemulsion formation can be attributed to an insufficient concentration of the surfactant. The surfactant, capable of solubilizing oil and water to form microemulsions, has been reported to exhibit a significantly enhanced ability for oil recovery.[30−33] Bera et al.[34] and Pal et al.[35] both demonstrated that the injection of microemulsions after water flooding can enhance oil recovery from 25 to 30% in sandpack flooding experiments. Javanbakht et al.[36] performed X-ray microtomography experiments and concluded that the microemulsions can solubilize nonaqueous phase liquids and achieve a higher incremental oil recovery of 13% than surfactant flooding after water flooding. However, these studies utilized prepared microemulsions as opposed to surfactants, which would form microemulsions in situ with the displaced oil phase. The oil phase within microemulsions has the capacity to solubilize oil in porous media. It could not be concluded that the high incremental oil recovery is attributed to surfactant or oil phase in microemulsions. Han et al.[37,38] proposed injection of SP or ASP solutions in a negative salinity gradient that could present the largest oil recovery factor by handling different types of microemulsion formation. Chen et al.[39] stated that the surfactant capable of in situ formation of middle phase microemulsions could enhance oil recovery by 11.25% compared to conventional ultralow interfacial tension ASP flooding in core flooding experiments. The surfactants employed in this study demonstrated notable emulsification capabilities. A significant presence of emulsions was noted in the effluent. The challenge lies in quantifying the extent of incremental oil recovery that can be directly linked to microemulsion formation or emulsion formation.

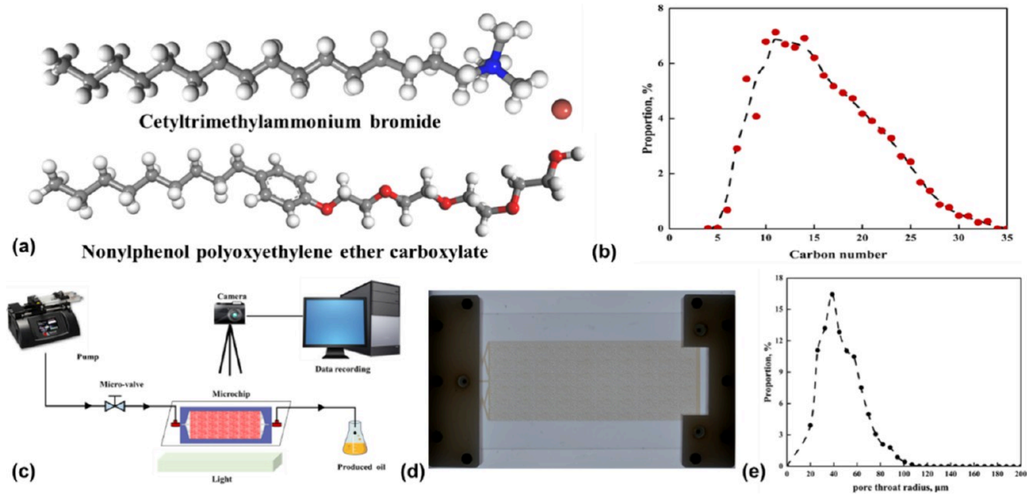

Figure 2. (a) Schematic diagram of the chemical structures of two surfactants. (b) Carbon number distribution of the simulated oil. (c) Schematic diagram of the micromodel experimental setup. (d) Real image of the micromodel. (e) Channel size distribution of the fabricated micromodel.

The aforementioned studies indicate that in addition to the commonly recognized mechanism of interfacial tension reduction, oil solubilization significantly contributes to the effectiveness of chemical enhanced oil recovery (EOR). However, the distinctions regarding oil mobilization within the porous medium by the solubilization effects induced by microemulsion formation or sweep area enlargement produced by emulsion formation remain ambiguous. Furthermore, chemically enhanced oil recovery is typically implemented during the high-water cut phase. The presence of an excess water phase within the pores will impede the interaction between the surfactant and oil phase. It may have different effects on the performance of oil mobilization by surfactants, which feature an IFT reduction or oil solubilization ability. The objective of this work is to compare the contributions of IFT reduction and oil solubilization on oil mobilization with and without water phase effects. A phase diagram was constructed to optimize the formulation of microemulsions, and the ultralow interfacial tension surfactant was evaluated utilizing the spinning drop interface tensiometer. The performance of two surfactants in oil mobilization, characterized by distinct displacement mechanisms, was evaluated through microchip and core flooding experiments, taking into consideration the effects of oil−water saturation. The anticipated results are expected to offer valuable insights into the predominant chemically enhanced oil recovery mechanisms across various oil mobilization stages, thereby facilitating surfactant screening tailored to specific reservoir conditions.

2. MATERIALS AND METHODS

2.1. Materials. Cetyltrimethylammonium bromide (CTAB) with a purity of 99% was purchased from Aladdin Biochemical Technology Co., Ltd. (Shanghai, China). Nonylphenol polyoxyethylene ether carboxylate (SS-231) with analytical purity was procured from Qingdao Changxing Hi-Tech Development Co., Ltd. (Qiangdao, China). It featured with good emulsification capability that have been reported in the work of Hou and Sheng.[40] Their chemical structures are shown in Figure 2a. Sodium chloride (NaCl) was procured from Tianjin Fuchen Chemical Reagent Co. Ltd. All experiments were utilized using deionized water generated by the BK-10B system from Qianjing Environmental Equipment Co. Ltd. Four *n*-alkanes, specifically hexane, decane, dodecane, and hexadecane, were utilized as the oil phase in the microemulsion-phase behavior tests. *N*-Butanol ($CH_3(CH_2)_3OH$, AR) was employed as a cosurfactant to enhance the properties of the microemulsion. Both compounds were procured from Macklin Biochemical Technology Co., Ltd. (Shanghai, China). In this study, we examined the influence of the complex components of crude oil on the phase behavior of microemulsions. The simulated oil utilized was a blend of crude oil collected from the Qinghai oilfield (PetroChina, China) and mineral oil from Beijing Yinghaiqing Industrial Lubricating Oil Co., LTD, prepared in a weight ratio of 1:1. The viscosity of mixed oil was 5 mPa·s at room temperature. The carbon number distribution of the simulated oil is shown in Figure 2b. Unless otherwise specified, the term "oil" throughout the manuscript refers to the simulated oil. The temperature remains stable at a consistent 297 K, reflecting the ambient conditions of the room.

2.2. Phase Behavior and Phase-Diagram Construction. Since there is no microemulsion formation when SS-231 mixed with alkanes or crude oil, CTAB mixed with *n*-butanol as a weight ratio of 2:3 acted as the surfactant in the phase behavior research. Our previous tests found that CTAB with *n*-butanol could form amounts of microemulsions when their mixed concentration was higher than 5 wt %. Therefore, the concentration of CTAB mixed with *n*-butanol of 5 wt % was used in this work. The surfactant solution was prepared by using a rotating stirrer with a rotation speed of 200 rpm. An equal volume of mineral oil and 5 wt % surfactant was mixed in a tube. The mixed solution was turned upside down 20 times and equilibrated at 297 K. Each phase volume was observed after fully equilibrium. The concentration of NaCl was varied from 1 to 9 wt % and kept surfactant concentration constant. Each microemulsion type and oil−water solubilization could be calculated from the phase volumes. Then, the optimal salinity was known from the intersection of the oil solubilization curve and water solubilization curve, which could be used for the following oil displacement experiments.

The simulated oil used in the oil displacement experiments featured a wide carbon number distribution. The phase diagram of different alkanes with surfactant could help understand the effect of the carbon number on microemulsion formation. The titration method was adopted for the construction of pseudoternary phase diagram.[41,42] The weight ratio of CTAB and *n*-butanol was fixed at 2:3 as one component in the pseudoternary phase diagram. The concentration of sodium chloride was fixed at 6 mg/L based on the salinity screening results from the phase behavior tests. The surfactant and brine were mixed with weight ratios of 1:9, 2:8, 3:7, 4:6, 5:5, 6:4,7:3, 8:2, and 9:1 and prepared at a temperature of 297 K. The oil phase including hexane, decane, dodecane, and hexadecane were separately added into the surfactant and brine-mixed solution with stirring. The point that the solution change from clear and transparent to cloudy corresponded to the transition point of the single-phase region to the multiphase region. These points were recorded for phase-diagram construction.

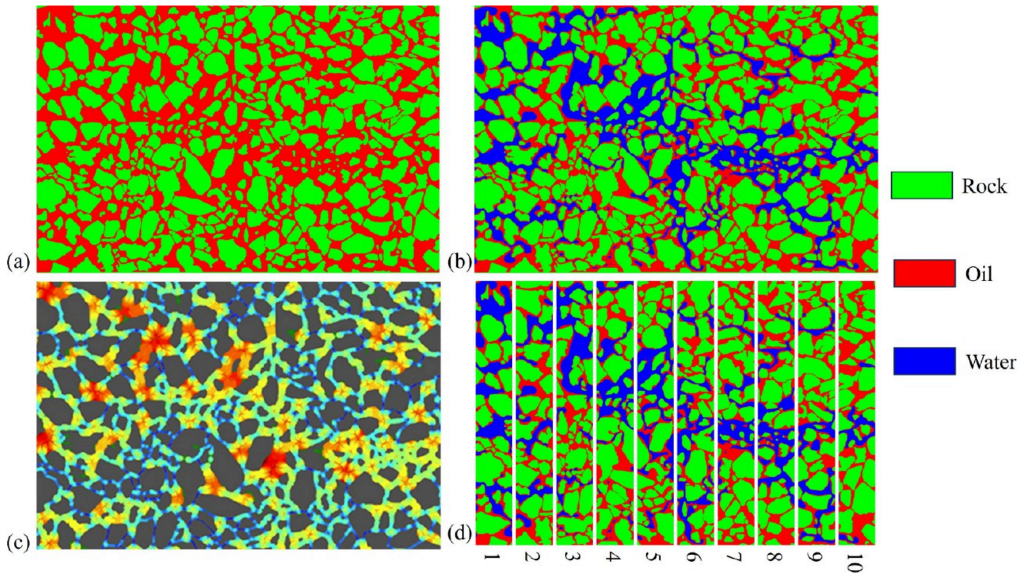

Figure 3. Schematic diagram of the recovery factor calculation. (a) Initial saturated oil image; (b) distribution of oil−water during displacement. (c) Method for dividing pore regions with different radii. (d) Calculation method for regional recovery factor.

2.3. Interfacial Tension Measurement. The surfactant solution containing 0.5 wt % SS-231 and 3 wt % NaCl was prepared for the IFT measurement. The IFT between surfactant and mineral oil, crude oil, and simulated oil were separately measurand using a Texas-500C spinning drop interface tensiometer (CNG company, US) with a rotation speed of 5000 rpm based on the Vonnegut equation. The IFT between CTAB with *n*-butanol and simulated oil failed in direct measurement using an interface tensiometer because their formed microemulsions have no obvious oil−water interface. Its value could be calculated by using the Huh equation. The Vonnegut equation and Huh equation are found in the Supporting Information.

2.4. Morphology of Emulsion and Microemulsion. Emulsions were formulated utilizing the surfactant SS-231 at a concentration of 0.5 wt %, combined with mineral oil in a volume ratio of 1:1. The blended solution underwent agitation with a homogenizer set at 100 rpm for a duration of 10 min. The emulsion's morphology was examined using a stereo zoom microscope (Zeiss-Discovery StereoV12). Micrographs of the microemulsion were acquired using a HITACHI Regulus8100 cryo-microscope, operating at 5 kV with a working distance of 6.1 mm, utilizing scanning electron microscopy techniques. For the acquisition of surface views of the middle phase microemulsion, 20 μL of the microemulsion was applied to a conductive carbon tape and subsequently subjected to freeze-drying. Prior to SEM imaging, the samples underwent a coating process with gold for 20 s utilizing an HITACHI MCI000 sputter coater. To validate the formation of the middle phase microemulsion, a Zetasizer-Nano-S dynamic light scattering (DLS) apparatus (Malvern Instruments Ltd., UK) was employed for measuring the size distribution.

2.5. Micromodel Design and Experimental Procedure. The investigation focused on oil mobilization through surfactants exhibiting ultralow interfacial tension and microemulsions, assessed via oil displacement within the micromodel. The configuration of the micromodel platform is illustrated in Figure 2c. The setup comprised a color camera (Sony α7RIII), a dual-channel constant-rate syringe pump (Harvard PHD Ultra 2000), a stereo zoom microscope (Zeiss-Discovery StereoV12), and a glass etching micromodel. The images were captured in real time by the camera, and the local zoom was distinctly observed through the microscope. The micromodel depicted in Figure 2d, measuring 100 × 36 mm, was constructed by replicating a smaller unit sized at 6 × 4 mm, which was designed in accordance with an image of the pore structure observed in the natural core. The etching depth of the glass micromodel measured 30 μm, with the channel size distribution ranging from 20 to 200 μm, as illustrated in Figure 2e.

Four experiments were designed to compare the oil mobilization performance of two surfactants, which separately featured oil solubilization and interfacial tension reduction characteristics by considering the effect of the aqueous phase. CTAB with *n*-butanol acted as the oil solubilization formulation agent, while SS-231 acted as the interfacial tension reduction formulation agent. Although the surfactant of SS-231 could achieve the ultralow IFT with oil even if the concentration was lower than 0.5 wt %, the concentration of two formulations was set at 5 wt % for the comparison. They were separately injected into the oil-saturated micromodel fore and after a water flood. Each stage was conducted at a constant flow rate of 1 μL/min until no color change was observed. Oil recovery was identified through the pixels changing of oil, water, and rock. In addition, the oil recoveries of each pore size and different positions of the micromodel could be achieved by performing the same operation on the specific pore radius space and different regions of the micromodel. Schematic diagram of above processing procedure is shown in Figure 3. The details could be referred our previously published work.[43,44]

2.6. Coreflood Experiments. Five outcrop cores, each with a diameter of 2.5 cm and a length of 30 cm, were utilized to validate the influence of two mechanisms previously examined in oil displacement experiments at the core scale. The cores exhibited an average porosity of 24.3% and an average permeability of 125 mD. The samples underwent vacuum treatment and were subsequently saturated with brine. Following this, oil was injected into the outcrop cores at a controlled flow rate ranging from 0.1 to 1 mL/min to establish the initial oil saturation levels. Five experimental schemes are presented below: One outcrop was subjected solely to water flooding for the purpose of comparison. The formulation agent for oil solubilization and the formulation agent for interfacial tension reduction, both at a concentration of 5 wt %, were injected separately into two distinct outcrop cores until oil production ceased. To elucidate the influence of the presence of water on the efficacy of two chemicals characterized by distinct oil mobilization mechanisms, one pore volume (PV) of each chemical was individually injected into two outcrop cores following a water flooding process until the water cut attained 80%. Subsequently, the postwater flooding process was carried out until there was no further oil production observed. The injection flow rates were established at 0.1 mL/min, reflecting the actual seepage velocity observed within the reservoir. The recorded injected pressures and effluent data were utilized for the calculation of the oil recovery.

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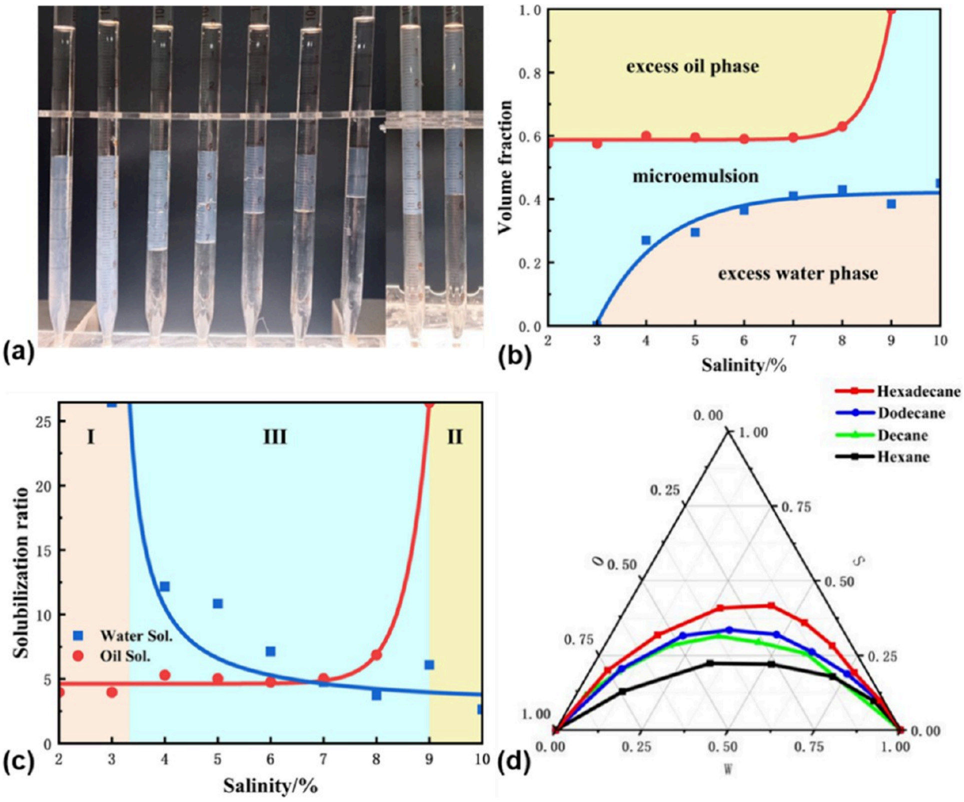

Figure 4. (a) Phase behavior images of CTAB mixed with *n*-butanol and mineral oil. (b) Relative volume fraction of three phases versus salinity. (c) Oil and water solubilization parameters versus salinity. (d) Pseudoternary phase diagrams of surfactant/brine/oil system at 297 K at fixed ratio of CTAB/*n*-butanol = 2:3 for different oil types.

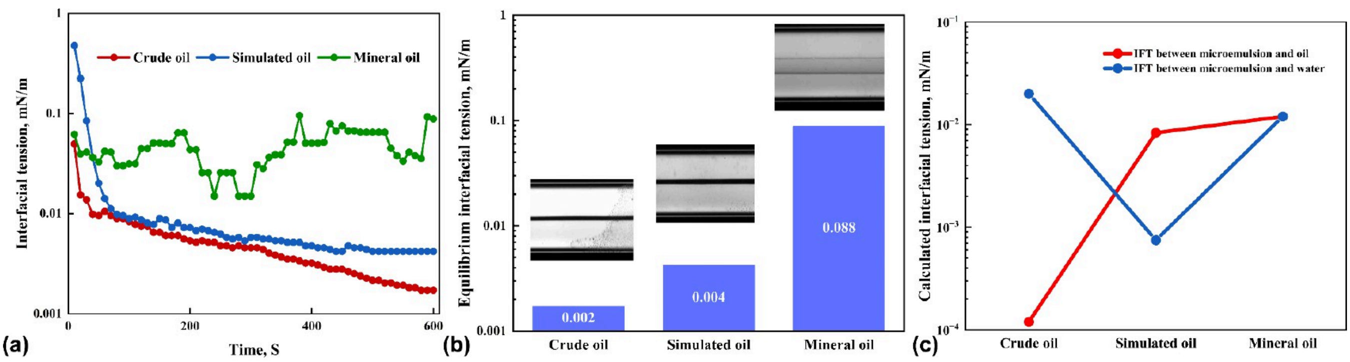

Figure 5. (a) Dynamic IFT curves between different oil and SS-231 surfactant using a spinning drop interface tensiometer. (b) Equilibrium IFT value and image between different oil and SS-231 surfactant. (c) Calculated IFT between microemulsion and oil or water phase using Huh equation for the microemulsion formular (CTAB mixed with *n*-butanol).

3. RESULTS AND DISCUSSION

3.1. Solubilization Parameters and Ternary Phase Diagram. The produced micelle will facilitate the solubilization of oil and water in conjunction with a solvent or cosurfactant, provided that the surfactant concentration exceeds the critical micelle concentration (CMC). The process results in the development of a microemulsion. The hydrophobic characteristics of the surfactant exhibited an increase in response to increased salinity levels. Consequently, the microemulsion of Winsor I type was initially observed at NaCl concentrations ranging from 2 to 3 wt %. It subsequently transitioned to the Winsor III type as the NaCl concentration increased from 4 to 8 wt %. It finally shifted to the Winsor II type when the concentration of NaCl was higher than 9 wt %, as shown in Figure 4a. The volumes of oil phase, water phase, and microemulsion versus salinity in Figure 4b presented that the salinity window for the system of CTAB/*n*-butanol and mineral oi was the concentration of NaCl from 4 to 8 wt %. The solubilization ratio intersection of the oil phase and water phase in Winsor III-type microemulsion in Figure 4c indicated that the optimal salinity for this system was at a NaCl concentration of 7 wt %. It has the equal oil and water solubilization ratio at the optimal salinity condition, which correspond to the lowest IFT between oil and water.

The study conducted involved a simulated oil with a broad carbon number distribution, leading to an investigation into how the carbon numbers of alkanes influence microemulsion formation through the construction of phase diagrams. The phase diagrams illustrated in Figure 4d delineate the conditions

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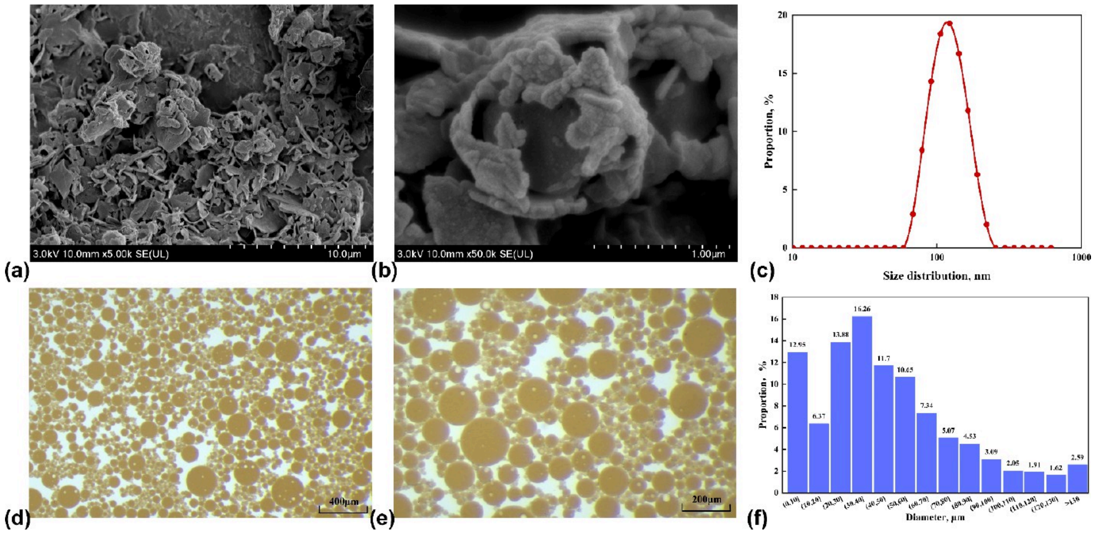

Figure 6. (a, b) Cryo-SEM images of the middle phase in the system of CTAB mixed with *n*-butanol and mineral oil in different magnifications; (c) size distribution of microemulsions tested by the dynamic light scattering apparatus; (d, e) morphology of prepared emulsions using SS-231 and simulated oil in different magnifications; (f) size distribution of emulsions by images statistics.

under which microemulsion formation occurs with a surfactant-mixed solvent, brine, and various alkanes. The regions enclosed by the envelope curve represent the multiphase zone, while the areas beyond delineate the single-phase zone. The single phase pertains to the microemulsion system. The ratios of each component can be determined from the phase diagrams, facilitating the screening process for microemulsion formulations. The analysis indicates that for the surfactant CTAB combined with *n*-butanol, the extent of the single-phase region diminishes as the carbon number increases. The findings demonstrated that this formulation is suitable for the simulated oil as it has the capability to create microemulsions with a range of alkanes possessing varying carbon chain lengths. The influence of salinity and surfactant concentration on the phase behavior of CTAB in conjunction with *n*-butanol and simulated oil is illustrated in Figure S1. The salinity at a NaCl concentration of 7 wt % and a surfactant concentration of 5 wt % resulted in the formation of distinct middle phase microemulsions, which were subsequently utilized in experiments concerning oil mobilization through microemulsions.

3.2. IFT between the Oil Phase and Surfactant Solution. The simulated oil was prepared by a mixture of crude oil and mineral oil. The IFT between three kinds of oil and SS-231 surfactant was measured to clarify the possible effect of oil component separation on oil mobilization. Dynamic IFT curves in Figure 5a showed that the IFT between the SS-231 surfactant and crude oil or simulated oil significantly decreased and then flattened with the test timing increasing. However, the IFT between the SS-231 surfactant and mineral oil throughout kept fluctuating around 10^{-2} mN/m. The equilibrium IFT values for simulated oil and crude oil in Figure 5b were both lower than 10^{-2} mN/m while it was 10^{-2} mN/m for the mineral oil. The heteroatom and nonhydrocarbon compounds in crude oil may have a stronger electrostatic attraction ability with the carboxyl groups of SS-231, which could flatten the oil−water interface and decrease IFT. The microemulsion formulation exhibits a strong solubilizing capacity with oil, which complicates the direct measurement of its interfacial tension with oil, as a distinct oil−water interface is not present. The Huh equation was employed for the interfacial tension calculation through the oil and water solubilization ratio. The computed interfacial tension encompasses the interfacial tension between microemulsions and oil, as well as the interfacial tension between microemulsions and water. All three oils have the potential to create a middle phase microemulsion when combined with a water phase consisting of CTAB and *n*-butanol. The interfacial tension measurements between the formed microemulsions and two phases exhibit significant differences. The detailed results are found in Figure 5c and Table S1. There are at least one phase's IFT that was lower than 10^{-3} mN/m via the calculation using the Huh equation for crude oil and simulated oil. For the mineral oil, the IFT between two phases and microemulsion are both around 10^{-2} mN/m because the brine used is optimal salinity based on the results of Section 3.1. The results of IFT measurement and calculation indicated that two formulars possessed similar IFT but different oil and water solubilization ability, which could be used for the comparison of oil mobilization preformation between two mechanisms of oil solubilization and IFT reduction.

3.3. Characteristic of Emulsions and Microemulsions. The ultralow IFT between simulated oil and SS-231 could result in emulsion formation, while the CTAB mixed with *n*-butanol could form microemulsion with the mineral oil. Their morphologies were separately observed by cryo-SEM and a microscope. Sponge-like structures of microemulsion in Figure 6a,b were in good agreement with other work reported typical morphology of middle phase microemulsion.[45,46] The sizes distribution of the middle phase solution tested by DLS in Figure 6c was from 60 to 300 nm, and its peak was concentrated around 150 nm, which again verified that there was the microemulsion generation using the CTAB mixed with *n*-butanol and mineral oil. The images of emulsions under the magnification of 50× and 100× in Figure 6d,e appear to be a regular sphere, and their sizes were more concentrated from 20 to 60 μm in Figure 6f. Both indicate that SS-231 featured a good emulsification ability with the simulated oil. Different properties of two surfactants on emulsion and microemulsion formation reconfirmed that they could be used for the

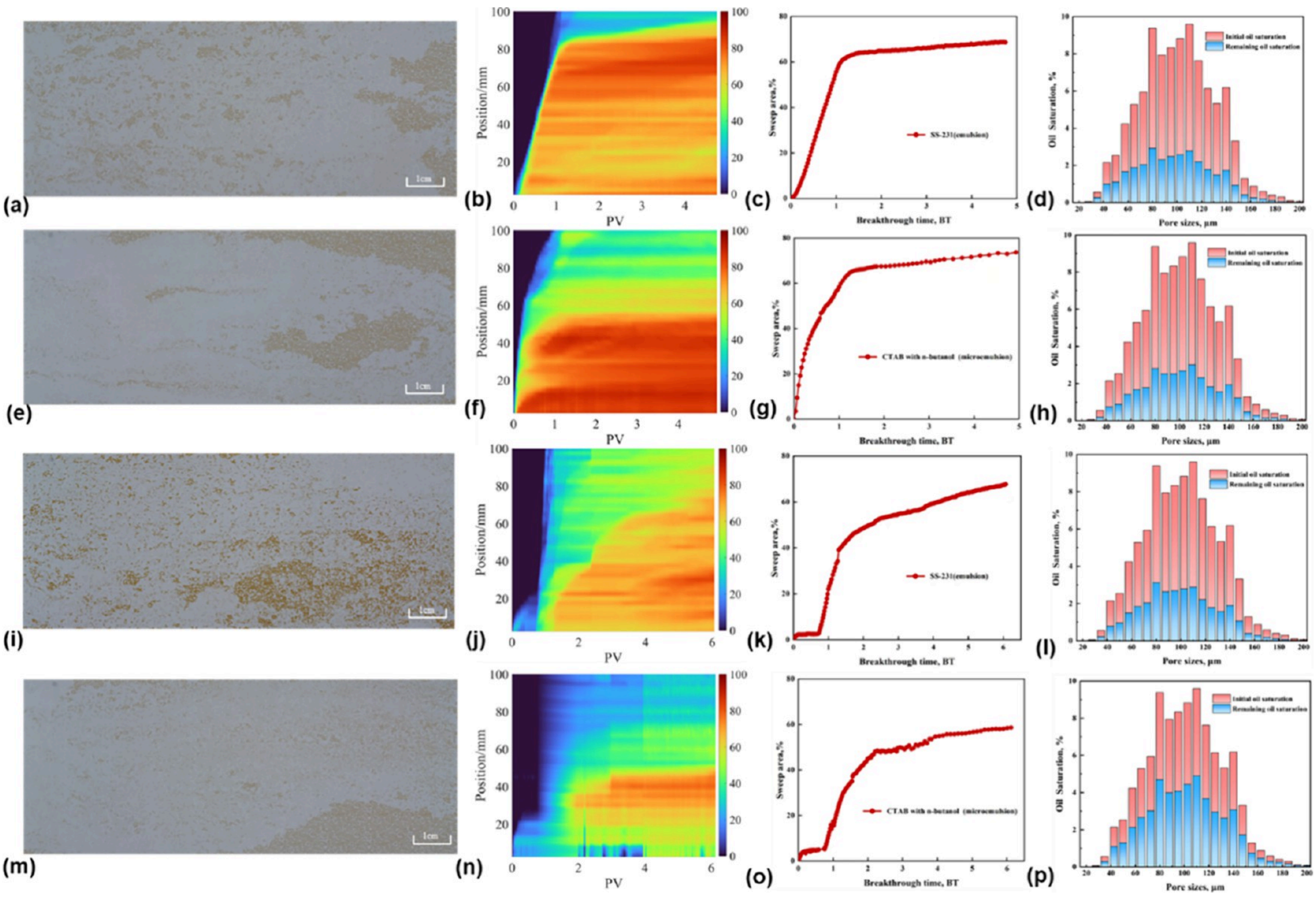

Figure 7. (a) Final remaining oil distribution, (b) the variation in recovery factor across various locations within the micromodel, (c) the curve representing swept areas, and (d) the remaining oil saturation corresponding to each pore size throughout the microscopic displacement process of surfactant SS-231 without prior water flooding. The meaning of panels (e)−(h) was consistent with the corresponding above images, illustrating the microscopic displacement process of the surfactant CTAB in conjunction with *n*-butanol, conducted without prior water flooding. Panels (i)−(l) convey the same concept, referring to the microscopic displacement process of surfactant SS-231 followed water flooding. (m−p) The microscopic displacement process of surfactant CTAB with *n*-butanol followed water flooding. The SS-231 exhibited ultralow interfacial tension and demonstrated effective emulsification capabilities. The combination of CTAB with *n*-butanol facilitated the formation of a microemulsion with the oil.

comparison of oil mobilization by solubilization and interfacial tension reduction.

3.4. Pore-Scale Comparison of Oil Mobilization by Solubilization and IFT Reduction. The surfactant-induced emulsification of oil leads to the formation of emulsions, while the solubilization of oil results in microemulsions, both of which are considered primary mechanisms for oil displacement. The prevailing mechanism under diverse reservoir conditions remains ambiguous and lacks a definitive conclusion. The surfactants SS-231 and CTAB, in conjunction with *n*-butanol, exhibited distinct interfacial properties that were utilized to compare the significance of two oil mobilization mechanisms through microscopic displacement experiments. Two surfactants were separately injected into the oil saturated micromodel before and after water flood. The four displacement process images are shown in Figures S2 and S3. The four displacement experimental results presented in Figure 7 indicate that the presence of the aqueous phase significantly influences the performance of the surfactants across various oil displacement mechanisms. The CTAB with *n*-butanol could get an ultimate swept area of 73.7% without water flooding in Figure 7g, which was 5% higher than that of SS-231 in Figure 7c. However, the results became different if surfactant was injected after water flooding. The ultimate swept area of SS-231 in Figure 7k was 67.4%, which was obviously higher than that of 58.7% of CTAB with *n*-butanol in Figure 7o. The findings demonstrated that oil solubilization played a crucial role in oil mobilization when solely the oil phase was present in the porous medium. Additionally, the presence of ultralow interfacial tension, coupled with effective emulsification properties, proved to be advantageous in scenarios where both oil and water coexisted within the pores.

The final remaining oil images and the variation in the recovery factor across different positions in the micromodel depicted in Figure 7 may assist in analyzing the underlying reasons. The remaining oils after SS-231 flooding in Figure 7a,i were more dispersed than that of CTAB with *n*-butanol in Figure 7i,m whether water flooding was conducted or not. There were more remaining oils in the front half of the micromodel after the SS-231 flooding and more oil leaving near the outlet of the micromodel after the CTAB with *n*-butanol flooding. The oil recovery factors of each position in four micromodels were calculated via the image analysis during the whole displacement process. In comparison to SS-231, the red color exhibited a higher concentration within the first half of the micromodel illustrated in Figure 7f during the CTAB

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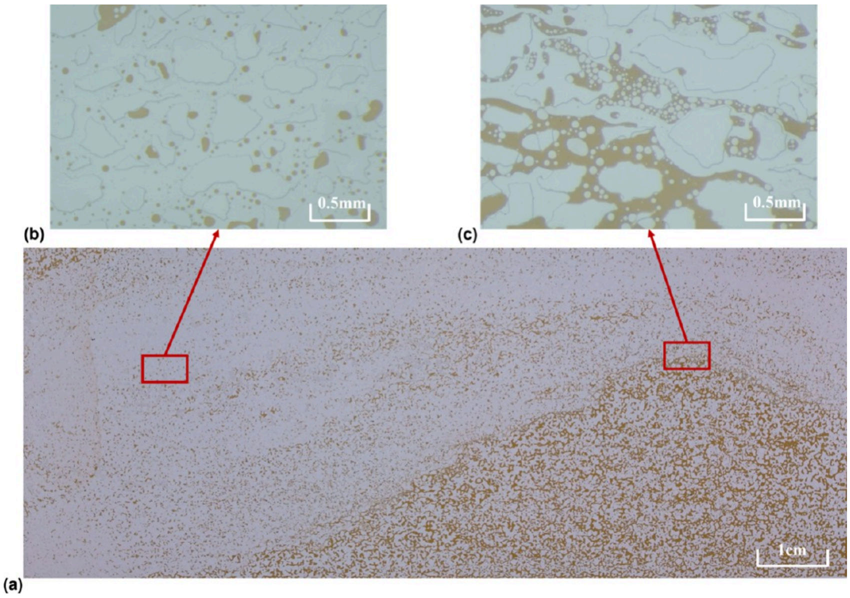

Figure 8. Microscopic displacement images in the SS-231 flooding process: (a) oil distribution in the whole microchip; (b) magnified images of the residual oil after displacement; (c) magnified images at the edge of oil and water during the displacement.

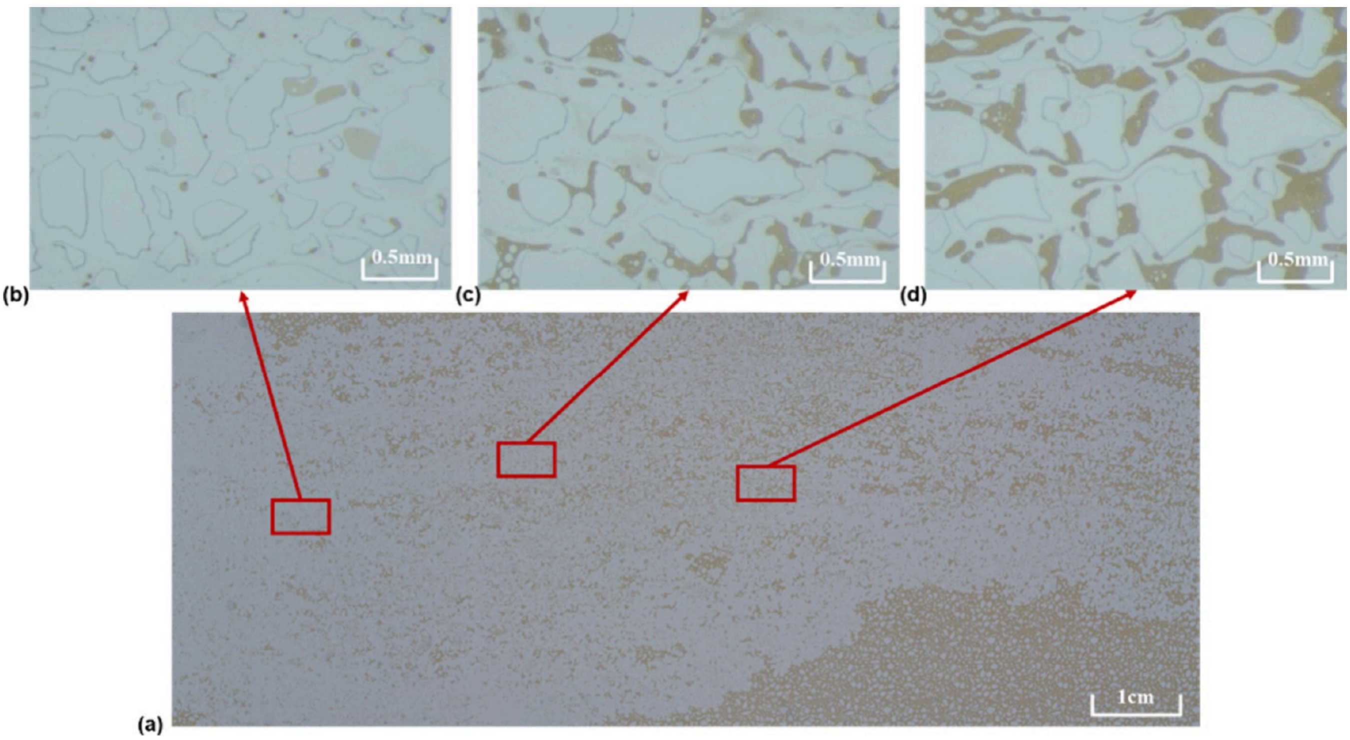

Figure 9. Microscopic displacement images in the CTAB with *n*-butanol flooding process. (a) Oil distribution in the whole microchip; (b) magnified images of the light-colored oil droplets after displacement; (c) light-colored oil films contiguous distribution. (d) Oil displaced in a filamentous form.

and *n*-butanol flooding process. There was a clear reduction observed in Figure 7n following the injection of CTAB with *n*-butanol subsequent to water flooding. However, SS-231 demonstrated the capability to maintain a relatively high efficiency in swept areas, as illustrated in Figure 7j. It indicated the excess aqueous phase hindered the role of the mechanism of oil solubilization. The pore-scale-remaining oil of four experiments counted in Figure 7 showed that each pore possessed the most remaining oil in the case of CTAB with *n*-butanol flooding after the water flooding. It was consistent with the above conclusion. The oil recovery from each pore shown in Figure S4 indicates that, in comparison to SS-231, the use of CTAB combined with *n*-butanol facilitates the mobilization of oil within smaller pores. However, there was no significant difference in the sizes of displaceable pores when surfactants were injected post water flooding. The microscopic displacement results demonstrated that the mechanism of oil solubilization requires the mass transfer of surfactant molecules. The complete realization of the oil solubilization

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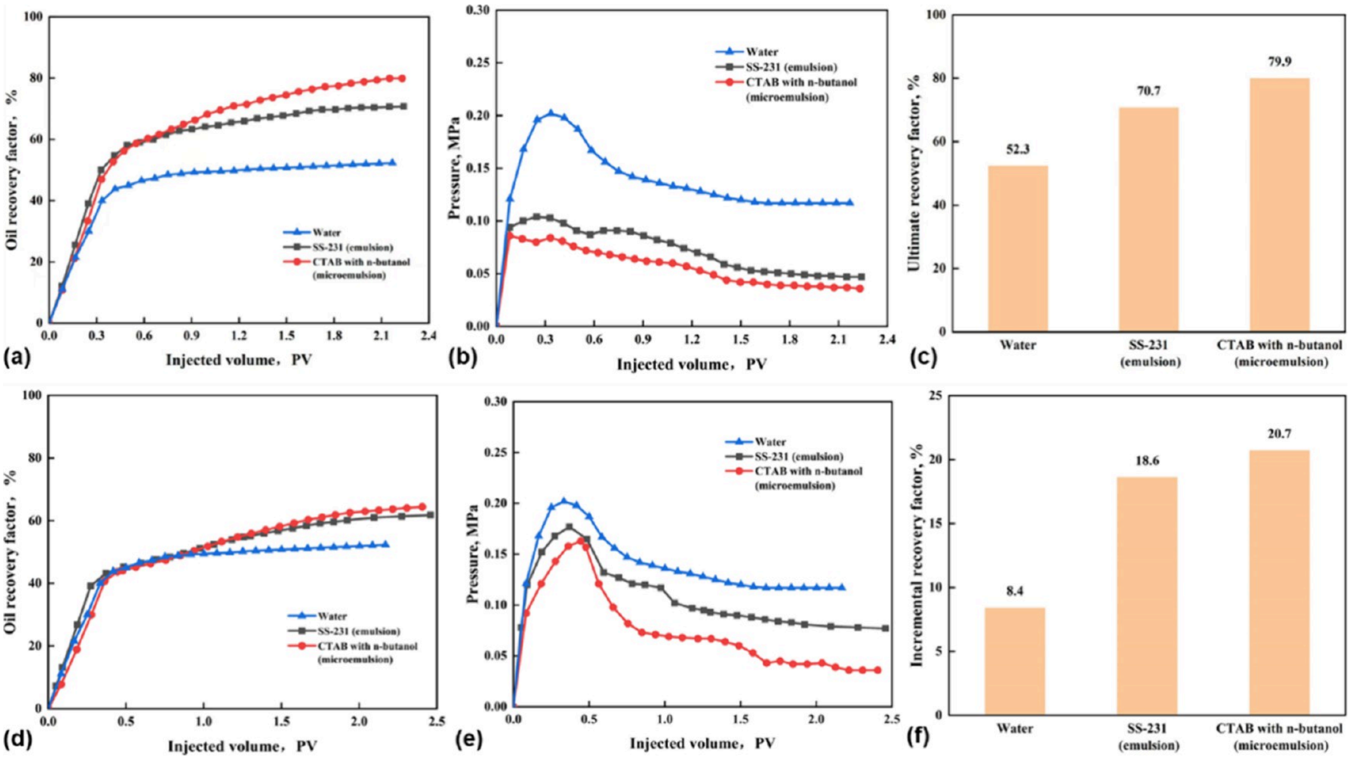

Figure 10. (a) Oil displacement efficiency curve, (b) injection pressure curve, and (c) the ultimate oil recovery factor of water flooding, SS-231 flooding, and CTAB with *n*-butanol flooding without prewater flooding. (d) Oil displacement efficiency curve, (e) injection pressure curve, and (f) the incremental oil recovery factor of the same three fluids as above but were separately injected into the cores after water flooding. The SS-231 could form emulsions with the oil, and CTAB with *n*-butanol could form microemulsion with the oil.

effect necessitates a high concentration of surfactants, presenting significant challenges for practical applications.

The magnified images of the oil mobilization process by two surfactants featured with different oil displacement mechanisms are separately shown in Figures 8 and 9. In the SS-231 flooding process, residual oil was distributed among the pores in the form of dispersed oil droplets after displacement, as shown in Figure 8b. There were many water-in-oil emulsions in Figure 8c when SS-231 just came into contact with the oil. It can be converted into the oil-in-water emulsion with the displacement proceeding. These emulsions also verified the SS-231 featured with good emulsification ability. The oil displacement process of CTAB with *n*-butanol in Figure 9 presents the oil mobilization phenomenon of microemulsions. The light-colored oil droplets in Figure 9b and contiguous oil films in Figure 9c indicated that the oil had been solubilized by the surfactant. Oil was prone to deform filamentous in Figure 9d, which meant that the IFT between oil and aqueous phase could reach ultralow. The oil solubilization and ultralow IFT jointly played important roles in the oil mobilization by microemulsions

3.5. Effect of Water Presence on the Performance of Two Chemicals Featured with Different Oil Mobilization Mechanisms. The performance of oil mobilization for the two surfactants, each characterized by interfacial tension reduction and oil solubilization, was evaluated through core flooding experiments to elucidate the primary mechanisms of oil displacement at the core scale. The experimental design adhered to the methodology established in the pore-scale oil displacement studies conducted within the micromodel framework. The oil displacement results presented in Figure 10a,c indicate that the CTAB system combined with *n*-butanol demonstrates a notable oil solubilization capability, achieving an ultimate oil recovery of 79.9%. This performance is significantly superior to those of SS-231 at 70.7% and water at 52.3%. In comparison to water flooding, the oil production during surfactant flooding, as illustrated in Figure 10a, continues to rise even after the injection of 0.3 pore volume (PV) of fluid. This observation suggests that surfactants have the potential to prolong the breakthrough of the displacing phase. The oil production of CTAB with *n*-butanol was significantly more than that of SS-231 after 0.9 PV of fluid injection, which resulted in an oil recovery difference of nearly 9%. The lowest injected pressure of CTAB with *n*-butanol in Figure 10b also verified that it featured good oil solubilization ability in the porous medium because the formed microemulsion could greatly reduce the flow resistance between oil and water. These results indicate that when only oil phase existed in porous media, the oil solubilization by surfactant was more effective than reducing IFT for improving oil recovery.

However, the difference of oil recovery caused by oil solubilization and IFT reduction gradually disappeared when there were two oil and water phases in the porous medium. Figure 10d shows the oil recovery factors of two surfactants further increased when they were injected after water flooding to the water cut reached 80%. Their ultimate recovery factors were both around 60%, which were significantly lower than those directly conducting surfactant flooding in Figure 10c. The difference of 8.4% in Figure 10f between the oil recovery factor at the water cut of 80% and ultimate water flooding recovery factor was selected as a comparison standard. The incremental oil recovery factors of SS-231 and CTAB with *n*-butanol were 18.6 and 20.7%, which indicated that two surfactants could enhance oil recovery even after water

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flooding. However, there was no remarkable difference between two surfactants. The injection pressures of two surfactants in Figure 10e were similar to surfactant flooding without water flooding in Figure 10b. The formed emulsions during the SS-231 injection made its injection pressure larger than that of CTAB with *n*-butanol but lower than that of water flooding because the ultralow IFT between oil and water weakened the effect of capillary force.

The oil displacement efficiency in the micromodels and cores both showed CTAB with *n*-butanol featured with good oil solubilization capability could acquire a higher oil recovery factor than that of SS-231 featured with ultralow IFT and good emulsification capability. However, this difference decreased when there were two oil and water phases in the porous medium, even the SS-231 possessed a higher incremental oil recovery after water flooding in the microscopic displacement experiments. It can be concluded that oil solubilization plays an important role for oil mobilization if there is no water existing, and emulsion formation becomes considerable for enhanced oil recovery at high water saturation. Many works have reported that compared to injecting surfactant after water flooding, the surfactant flooding without water flooding was featured with a higher oil recovery factor.[26,47] It indicated that the aqueous phase will affect oil mobilization, which was in agreement with this finding.

Our work could provide guidance for surfactant selection. The surfactant featured with the good emulsification capability was preferred at the high-water cut stage, while surfactants featured with the high oil−water solubilization capacity could be a good candidate when the remaining oil saturation was high in the reservoir. Surfactant properties should be carefully evaluated for high-salinity reservoirs. The divalent ion may cause surfactant precipitation, make Winsor III microemulsion transfer to Winsor II microemulsion, and decrease the stability of emulsions. The effect of these factors on surfactant selection should be further investigated in the followed work.

4. CONCLUSIONS

In conclusion, the comparison of two mechanisms—surfactant solubilization of crude oil and the formation of emulsions from emulsified oil—on oil mobilization was conducted through micromodel and core flooding experiments. Two surfactants were identified that can achieve ultralow interfacial tension, each exhibiting distinct mechanisms that were thoroughly screened and evaluated. The combination of CTAB and *n*-butanol has the potential to create microemulsions with alkanes that exhibit a broad range of carbon number distributions. The observation of middle phase microemulsions was conducted by combining CTAB with *n*-butanol and mineral oil, specifically at NaCl concentrations ranging from 4 to 8 wt %. The optimal salinity of 6 wt % NaCl brine was utilized for the preparation of the surfactant solution in the oil displacement experiments. Surfactant SS-231 did not succeed in forming microemulsions; however, it demonstrated effective emulsification properties with simulated oil. The morphology analysis confirmed that the combination of CTAB with *n*-butanol is capable of producing nanosized microemulsions when mixed with mineral oil, resulting in emulsions with dimensions ranging from 20 to 60 μm for SS-231 emulsified oil.

The use of CTAB in conjunction with *n*-butanol resulted in an ultimate swept area of 73.7% within the micromodel, achieved without the necessity for prewater flooding, surpassing the performance of SS-231 by 5%. However, the swept area was found to be 8.7% lower than that of SS-231 when surfactants were introduced following water flooding. The core flooding experiments yielded comparable results; however, the oil recovery factor for CTAB with *n*-butanol was observed to be 2.1% greater than that of SS-231 in the context of surfactant injection following water flooding. The experiments conducted at both the pore scale and the core scale demonstrated that the oil solubilization mechanism functioned effectively in the presence of only the oil phase within the porous medium. Conversely, the formation of emulsions became increasingly significant when both oil and water phases were present in the porous medium. The oil displacement in the two-dimensional micromodels reflected the swept volume enlargement ability of injection fluid, and the one-dimensional core flooding presented the displacement efficiency improvement ability. The results indicated that the mechanism of oil solubilization played a vital role in oil displacement efficiency improvement, and emulsification was more favorable for the swept volume enlargement.

This study revealed the significance of oil solubilization and interfacial tension reduction in facilitating oil mobilization across various application scenarios. The rationale behind the selection of ultralow interfacial tension as a criterion for surfactant screening by numerous pilots can be attributed to the significant oil recovery factors observed from the pore-scale oil displacement experiments conducted in this study. This study highlights that the mass transfer efficiency of surfactants at elevated water cuts significantly influences the oil solubilization potential, an aspect that must be considered during the surfactant selection process. Compared with oil solubilization, the role of emulsification became more important at the high-water cut stage. The emulsified capability should be a considerable property for surfactant screening toward enhanced oil recovery for high-water-cut reservoirs.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.langmuir.5c04342.

Interfacial tension calculation, phase behavior images, and pore-scale oil mobilization images (PDF)

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Notes

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